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RESEARCH ARTICLE

Impact of hyperspectral reconstruction techniques on the quantitative inversion of rice physiological parameters: A case study using the MST++ model

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Highlights

- A comprehensive evaluation of the MST++ hyperspectral reconstruction model and its impact on the inversion accuracy of rice physiological parameters.
- A systematic comparison of the performance of various regression algorithms on the reconstructed hyperspectral data on the inversion.
- Insights into the potential of hyperspectral reconstruction technology in agricultural remote sensing applications and recommendations for future development to enhance its applicability.

Abstract

Quantitative inversion is a major topic in remote sensing science. The development of visible light-based hyperspectral reconstruction techniques has opened novel prospects for low-cost, high-precision remote sensing inversion in agriculture. The aim of this study was to assess the effectiveness of hyperspectral reconstruction technology in agricultural remote sensing applications. Hyperspectral images were reconstructed using the MST++ hyperspectral reconstruction model and compared with the original visible light images in terms of their correlations with physiological parameters, the accuracy of single-feature modeling, and the accuracy of combined feature modeling. The results showed that compared to the visible light image, the reconstructed data exhibited a stronger

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correlation with the measured physiological parameters, and the accuracy was improved for both the single feature and combined feature inversion modes. However, compared to multispectral sensors, hyperspectral reconstruction provided limited improvement of the inversion model accuracy. The results suggest that for physiological parameters that are not easy to observe directly, deep mining of features in visible light data through hyperspectral reconstruction technology can improve the accuracy of the inversion model. However, appropriate feature selection and simple models are more suitable for the remote sensing inversion task of traditional agronomic plot experiments. To strengthen the application of hyperspectral reconstruction technology in agricultural remote sensing, further development is necessary with broader wavelength ranges and more diverse agricultural scenarios.

Keywords: multistage spectral-wise transformer, hyperspectral reconstruction, rice, dry matter content, height

1. Introduction

Remote sensing technology has played an increasingly important role in agricultural research over the past few decades, and it now offers a wide range of applications, including crop monitoring, yield prediction, disaster assessment, pest monitoring, and precision agriculture (Albergel *et al.* 2013; Chen *et al.* 2020; Weiss *et al.* 2020; Xu *et al.* 2020; Danner *et al.* 2021; Yang *et al.* 2021). Among various remote sensing approaches, visible light and multispectral imaging have been extensively used for agricultural remote sensing inversion (Yao *et al.* 2014; Tahir *et al.* 2018; Xie *et al.* 2019; Yi *et al.* 2019; Berger *et al.* 2020; Pei *et al.* 2023). However, the accuracy of these techniques is somewhat limited due to their low spectral resolution (Fitzgerald *et al.* 2010; Berger *et al.* 2018; Qi *et al.* 2020). On the other hand, hyperspectral imaging systems provide higher spectral resolution, but they are generally expensive and have limited applicability in large-scale agricultural monitoring (Liu *et al.* 2021; Elmanawy *et al.* 2022; Sun *et al.* 2022).

The development of visible light-based hyperspectral reconstruction techniques has opened new possibilities for low-cost, high-precision remote sensing applications in agriculture (Liu *et al.* 2022; Sousa *et al.* 2022; Gong *et al.* 2023). Hyperspectral reconstruction algorithms are used to estimate hyperspectral images from visible light or multispectral data, with the use of machine learning and optimization techniques to recover the high-dimensional spectral information lost during the imaging process (Fu *et al.* 2022; Yulong *et al.* 2022). This advancement has the potential to significantly improve the accuracy of agricultural remote sensing inversion while maintaining its affordability and accessibility for a wide range of users.

Currently, hyperspectral reconstruction techniques are being successfully applied in monitoring water quality and plant phenotyping traits (Yulong *et al.* 2022; Zhao *et al.* 2022). These techniques have also been used with great success in classification tasks using satellite remote sensing (Deng *et al.* 2021). However, their applicability in remote sensing inversion, particularly in agricultural fields, has not been extensively explored. Therefore, the potential applications of hyperspectral reconstruction techniques in agricultural remote sensing inversion need to be explored to identify the challenges and opportunities associated with their integration.

This study explores the potential of hyperspectral reconstruction technology in the inversion modeling of rice physiological parameters, which is an important topic in remote sensing and agricultural science (Chen *et al.* 2020; Tian *et al.* 2021). Despite many studies on the inversion modeling of various rice physiological parameters (Wang *et al.* 2022; Zhu *et al.* 2022; Liao *et al.* 2023; Liu *et al.* 2023; Yu *et al.* 2023), the application of hyperspectral reconstruction in rice has not been explored in previous research. Investigating the application prospects of hyperspectral reconstruction technology in the inversion modeling of rice physiological parameters can help to establish low-cost and efficient rice physiological parameter inversion modeling schemes.

To explore the application prospects of hyperspectral reconstruction techniques in the quantitative inversion of crop physiological parameters, this study focused on the Multistage Spectral-wise Transformer (MST++) model (Cai *et al.* 2022a, b), which has demonstrated superior performance in the field of hyperspectral reconstruction and won the First Place in the New Trends in Image Restoration and Enhancement workshop (NTIRE) 2022 Spectral Reconstruction Challenge. The purpose of this study was to develop inversion models for various rice physiological parameters, including the leaf, stem, panicle, and above-ground dry matter contents, plant height, SPAD value, and lodging resistance. By investigating the impact of hyperspectral reconstruction on inversion model accuracy from multiple dimensions (i.e., data and algorithms), we hope to provide valuable insights into

the potential benefits and challenges associated with integrating these techniques into agricultural remote sensing inversion workflows.

2. Materials and methods

In this study, data for the above-ground biomass content, plant height, SPAD value and bending resistance force were obtained by designing multi-variety and multi-nitrogen treatment rice experiments (Fig. 1). Remote sensing images were collected using the unmanned aerial vehicle (UAV) system (Fig. 2-A), and hyperspectral images with higher spectral resolution were obtained using hyperspectral reconstruction technology (Fig. 2-B). Six machine learning models were employed to establish inversion models with texture features and reflectance extracted from remote sensing images and the physiological parameters of rice obtained through field experiments (Fig. 2-C). Whether the hyperspectral reconstruction model used in this study could improve the inversion of rice physiological parameters was then determined by comparing the accuracy of the inversion model constructed from remote sensing images before and after reconstruction.

2.1. Experimental design

The experiment included two planting seasons in 2021 with four nitrogen treatments (N1 to N4, representing total nitrogen application amounts of 0, 75, 105, and 135 kg ha^{-1}, respectively) and eight rice varieties (H1 to H8, representing Mei Zhanxiang 2, Xiangyaxing, 19 Xiang, Ruanhua, Qingxiangyou 033, Nanjingxiang, Er Guangxiang 3, and Li Xiangzhan, respectively). The experimental field was located in Guangzhou, Guangdong Province, China (23.39°N, 113.44°E), in an area with a subtropical marine monsoon climate. The general layout of the experiment is shown in Fig. 1. Plot irrigation was implemented using clean water. Nitrogen (N) fertilizer, in the form of urea, was manually applied in split doses during the basal, tillering, and heading stages, following a distribution pattern of 50, 30, and 20%, respectively.

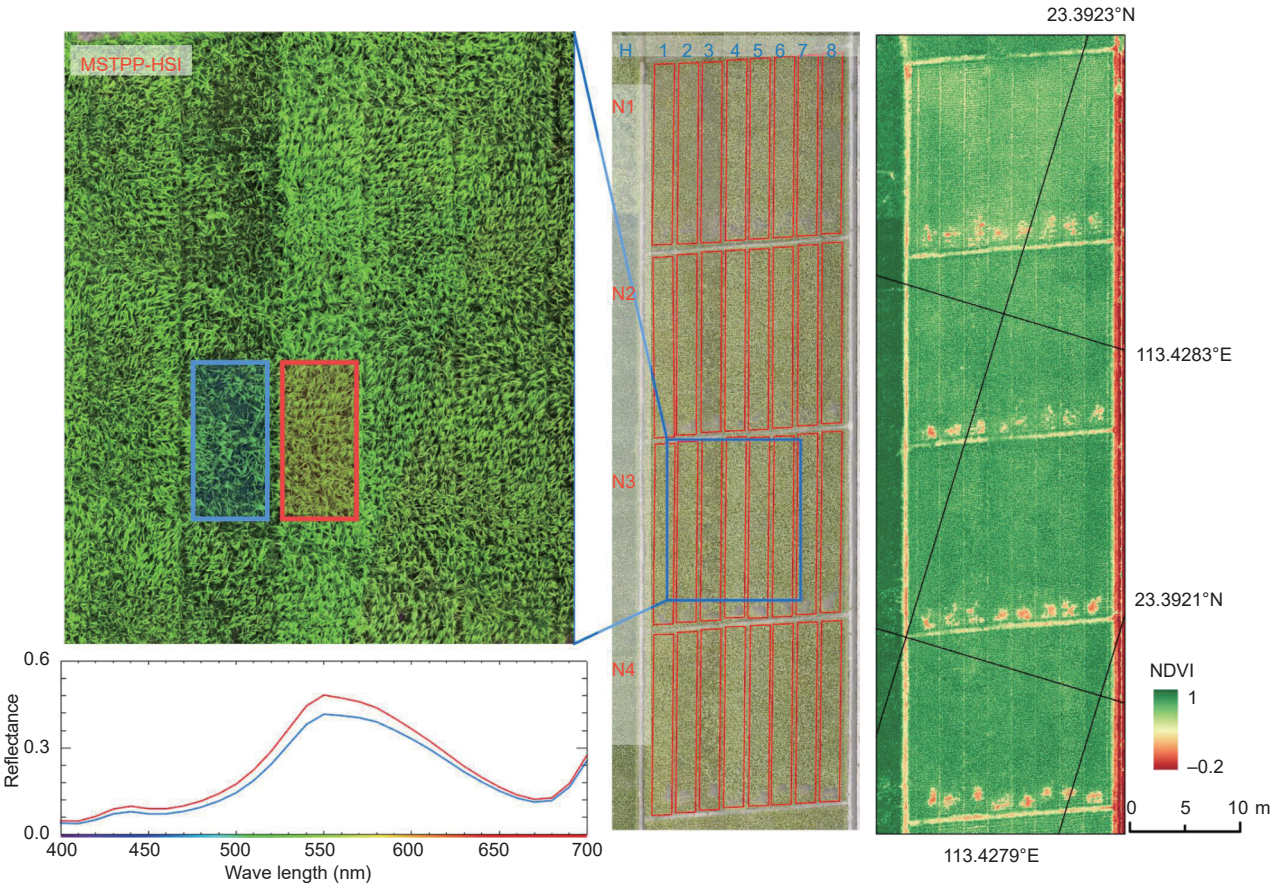


Fig. 1 Overview of the study area and illustration of the hyperspectral reconstruction results. The location of the experimental area and its overall layout are shown. Multistage spectral-wise transformer hyperspectral image (MSTPP-HSI) is the reconstructed hyperspectral image, and the spectral curves of the selected area are shown.

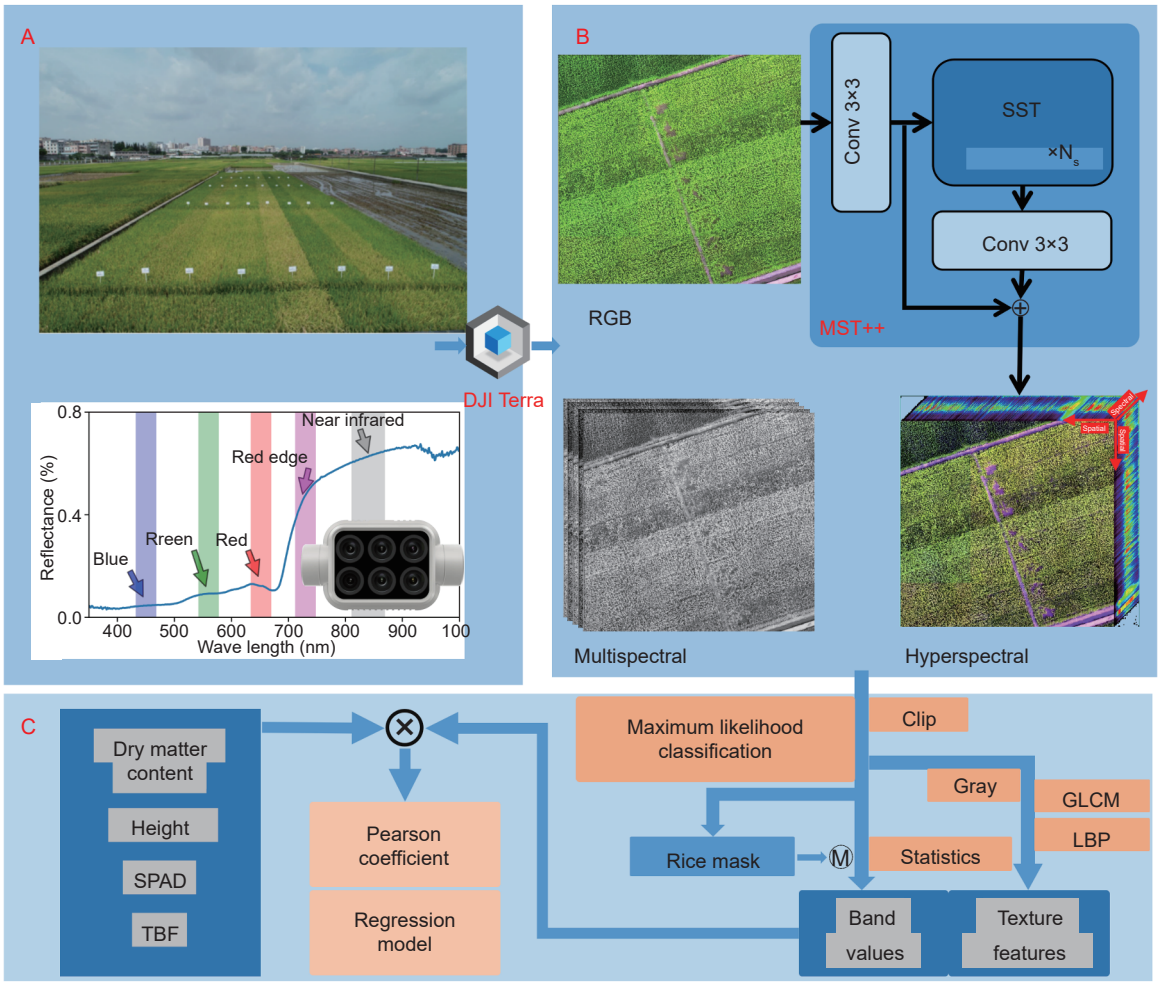


Fig. 2 Technical route. A, sensor information and experimental field used for data collection. B, preprocessing and hyperspectral reconstruction of the remote sensing data. C, data processing and analysis workflow. MST++, multistage spectral-wise transformer (MST++); SST, spectral-wise transformer; TBF, bending resistance force; GLCM, gray-level co-occurrence matrix; LBP, local binary patterns; RGB, visible light image.

The base fertilizer was administered 1–2 days prior to transplantation, while the tiller fertilizer was applied 5 days post-transplantation. The basal fertilizers, consisting of phosphorus (60 kg ha^{-1}) as calcium superphosphate and potassium (120 kg ha^{-1}) as potassium chloride, were spread and mixed into the topsoil through rotary tillage 1 day before sowing. Best management practices for high-yield fields were followed for water management, disease control, pest control, and weed control.

2.2. Data collection

Physiological parameter measurements The early rice growth period extends from March to July, while the late rice growth period extends from July to September. During the full heading stage of both early and late rice, data were manually collected for four types of physiological indices, including above-ground biomass, plant height, SPAD value, and bending resistance force. The above-ground biomass was further categorized into four components: leaf, stem, ear, and total above-ground biomass dry matter content (DMC).

From each experimental field, five rice plants were randomly selected from three regions and their leaf area, height, SPAD value and above-ground biomass content were measured (Table 1). The DMC was calculated as the above-ground biomass per unit area according to eq. (1). The SPAD value readings of 15 rice plants were collected in each experimental area using a SPAD meter (SPAD 502 Plus; Minolta Camera Co., Osaka, Japan). The lodging resistance characteristics of the population were measured by determining the bending resistance force (TBF) of 30 rice plants in each experimental field using a stem strength meter (YYD-1; Zhejiang Top

Instrument Co. Ltd., Hangzhou, China) (Liu *et al.* 2016) (Fig. 3). All physiological parameters were represented by their mean values for each experimental field.

$$DMC_p = \frac{\sum_{m=1}^{3}\sum_{n=1}^{5} DMC_p}{S \times m \times n} \times 10^8 \quad (1)$$

where *DMC* represents dry matter content per unit area of each experimental treatment area in units of tons per hectare (t ha^{-1}). *DM* represents dry matter weight in grams (g). The symbol *p* represents the part of rice, *m* and *n* represent the number of replicates in each experimental treatment and the number of samples in each replicate, respectively. In this study, the replicates of three areas were randomly collected for each experimental treatment, and five samples were randomly collected for each replicate. *S* is the planting area of each rice plant in square centimeters (cm^2).

Remote sensing data collection and preprocessing This study employed a DJI Phantom 4 multispectral drone (DJI, Guangdong, China) for remote sensing data collection. The platform has five single-channel sensors (blue, (450±16) nm; green, (560±16) nm; red, (650±16) nm; red edge, (730±16) nm; near-infrared, (840±26) nm) and a visible light (RGB) sensor, and is equipped with a light sensor to capture the real-time solar radiation intensity to ensure data consistency. The data were collected on June 20, 2021, and September 28, 2021, at noon when there was no cloud interference, at a flight height of 25 m. A DJI GS PRO ground station was used for task planning and data collection, with a flight direction and lateral overlap of 70%. The drone images were stitched using DJI Terra. In the following text, P4M, P4M-MS, and P4M-RGB are used to represent all the remote sensing images acquired by the Phantom 4 multispectral (P4M) drone, the multispectral images, and the visible light images, respectively.

2.3. Data processing

Hyperspectral reconstruction The reconstruction model adopted MST++ and its open-source weights

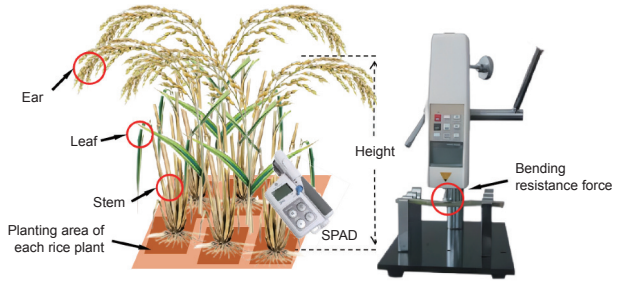


Fig. 3 Schematic diagram of physiological parameter measurement.

Table 1 Data overview

Descriptive statistics	Dry matter content (t ha^{-1})				Height (m)	SPAD	TBF (N)[1)]
	Stem	Leaf	Ear	Total			
Count	64	64	64	64	64	64	64
Mean	6.80	1.85	1.71	10.37	1.01	33.70	9.11
SD	0.87	0.26	0.22	1.14	0.20	4.00	1.63

[1)] TBF, the bending resistance force.

(https://github.com/caiyuanhao1998/MST-plus-plus) (Cai *et al.* 2022), which were based on the spatially sparse but spectrally self-similar nature of the hyperspectral image (HSI), to construct the basic unit of the spectral-wise attention block (SAB). Then SABs were used to build the single-stage spectral-wise transformer (SST), which addressed the limitations of convolutional neural network (CNN)-based methods in capturing long-range dependencies and self-similarity. In this study, Python 3.6 and the Rasterio library were used to perform hyperspectral reconstruction inference on the stitched visible light drone image, while maintaining the same spatial reference as the input image. A hyperspectral image with 31 channels, a wavelength range of 400 to 700 nm, and a bandwidth of 10 nm was obtained after reconstruction, and labeled as MSTPP-HSI. In the following text, MSTPP is used to represent the image obtained after hyperspectral reconstruction.

Background removal To obtain spectral data that were as accurate as possible, the influence of soil on the vegetation needed to be removed, which helps reduce the impact of background content on the region of interest (Yao *et al.* 2014). Through manual judgment and supervised learning, the ENVI5.6 classification tool was used to obtain a rice mask file using the maximum likelihood method. The classification accuracy of rice reached 99.25% using the Confusion Matrix tool, and the before-and-after classification results are shown in Fig. 4. Considering that the mask file would affect the texture features of the original image, the mask was only used for band value extraction but not for texture feature calculation.

2.4. Feature extraction

This study primarily used the band values and texture features of the images in different bands as the parameters for inversion. The boundaries of 32 experimental regions were established through manual drawing, and Python 3.6 was used to calculate the mean, maximum, minimum, and variance of each band in each experimental area as the feature vector of the band value.

This study used three typical texture feature extraction algorithms, gray-level co-occurrence matrix (GLCM), local

binary patterns (LBP), and image entropy, to extract the texture features of the remote sensing images before and after reconstruction. Since the texture features of each band have a high degree of similarity, only the mean of the images before and after reconstruction was considered for texture feature extraction.

GLCM GLCM is a statistical model used to describe the spatial distribution of gray levels in an image (Haralick *et al.* 1973). For the image cut by the experimental area boundary, this method first divides the image into several small areas (pixel pairs), then counts the number of times each pixel pair appears in the specified direction to obtain a matrix, the GLCM (eq. (2)). Each element of the matrix represents the number of times the two pixel values appear in the specified direction, thus describing the spatial distribution of gray levels. The GLCM features and their interpretation are given in Table 2.

$$\mathrm{GLCM}(i,j)=\sum_{x=1}^{N}\sum_{y=1}^{M}\begin{cases}1 & \text{if } I(x,y)=i \text{ and } I(x+\Delta x, y+\Delta y)=j\\ 0 & \text{otherwise}\end{cases} \quad (2)$$

where $I(x,y)$ represents the gray value at position (x,y) of the gray image, N and M represent the width and height of the image, respectively, Δx and Δy represent the distance and direction between pixels where the gray value is x and y.

The GLCM is composed of statistical data for the frequency of pixel pairs and reflects the patterns and rules of the gray levels of adjacent pixels. The features of the GLCM were calculated using the feature.greycomatrix and greycoprops functions from the Python skimage library. The mean value of each feature in each direction in the test area was recorded as the corresponding GLCM feature vector of that test area.

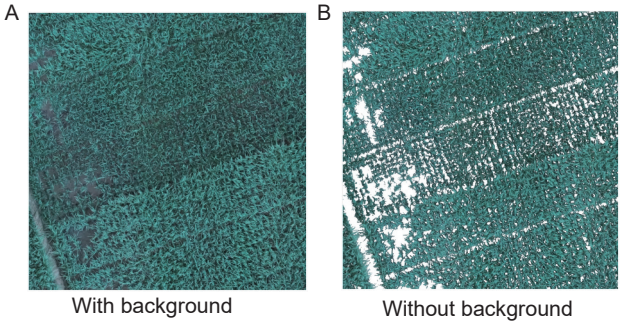


Fig. 4 Background removal.

LBP LBP is a feature extraction method used in image analysis, primarily for texture recognition. LBP compares each pixel to its surrounding pixels and converts the binary number into a new value for that pixel (Ojala *et al.* 1994, 1996). By performing this operation on the image after cropping the test area boundary and converting the resulting binary string into a decimal number, a feature vector that describes the texture of the image can be generated. In LBP, two parameters, P and R, can be used to control the number of pixels being compared and the size of the comparison radius. LBP can be applied to various image types, including gray-scale and color images. Its formula is as follows:

$$LBP_{P,R}=\sum_{P=0}^{P-1}s(g_P-g_c)2^P \quad (3)$$

where P is the number of samples in the neighborhood; R is the radius of the neighborhood; g_P is the gray level of the sample; g_c is the gray level of the center pixel; and s is the sign function, which is 1 when $g_P \geq g_c$ and 0 otherwise.

The LBP feature map of the test area was calculated using the local_binary_pattern function from the Python skimage library. The mean and standard deviation of the features in the test area were used as the LBP feature vector of the test area.

Image entropy The entropy of an image reflects the complexity of the pixel value distribution and represents the randomness of the information in the image. Its calculation involves using a histogram to estimate the proportion of each pixel value and then summing those proportions in the entropy calculation eq. to obtain the image entropy.

$$Entropy=-\sum_{i=1}^{L}P_i\log_2(P_i) \quad (4)$$

where L is the number of gray levels, and P_i is the

Table 2 Overview of gray-level co-occurrence matrix (GLCM) features and their interpretation

Feature	Description	Interpretation
Contrast	Measure the size of the gray level contrast in the image	A larger value indicates a stronger contrast
Correlation	Capture the correlation of gray levels in the spatial distribution of the image	A value closer to 1 implies a stronger correlation
Energy	Quantifie the probability of all gray levels appearing in the image	A larger value suggests a clearer image
Angular second moment (ASM)	Represent the average probability of all gray levels in the image	A larger value indicates a clearer image
Homogeneity	Evaluate the consistency of the gray level distribution in the image	A larger value signifies a more uniform distribution
Dissimilarity	Assesse the difference between gray levels in the image	A larger value reveals a more pronounced difference

probability of the *i*th gray level appearing in the image.

In this study, the image after cropping the test area boundary was binarized using the Otsu threshold algorithm, which divided the pixels in the image into two categories: black and white pixels. Then, the entropy of the image was calculated based on the distribution of the binarized pixel values.

2.5. Correlation assessment

The Pearson coefficient was used to assess the correlations between features and physiological parameters using the following eq.:

$$r_{xy}=\frac{\sum_{i=1}^{n}(x_i-\overline{x})(y_i-\overline{y})}{\sqrt{\sum_{i=1}^{n}(x_i-\overline{x})^2}\sqrt{\sum_{i=1}^{n}(y_i-\overline{y})^2}} \quad (5)$$

where x and y are the values of two variables, $\overline{x}$ and $\overline{y}$ are their means, respectively, and n is the sample size. The value range of r_{xy} is [–1, 1], and r_{xy}=1 indicates a completely positive linear relationship between the two variables, r_{xy}=–1 indicates a completely negative linear relationship between the two variables, and r_{xy}=0 indicates no linear relationship between the two variables.

2.6. Model construction and evaluation

To comprehensively assess the impact of the hyperspectral reconstruction model on inversion accuracy, this study employed six machine learning models of linear regression, polynomial regression, support vector machine regression (SVM), decision tree regression (DT), linear neural network regression (LNN), and multi-layer perceptron regression (MLP), all implemented in Python. The linear neural network used a Dense layer for linear transformation. The multi-layer perceptron featured three hidden layers, and a dropout layer linked to the output fully connected layer. Both networks use the mean squared error (MSE) as the loss function, mean absolute error (MAE) as the evaluation index, stochastic gradient descent (SGD) as the optimizer, and L2 regularization to prevent overfitting.

Five-fold cross-validation was applied to partition the dataset into training and test sets (Fig. 5). For the linear neural network and multi-layer perceptron, 10% of the training set was designated as the validation set for model evaluation during the training process. For the final model prediction evaluation, the coefficient of determination (R^2) and root mean square error (RMSE) were calculated for the test set.

$$R^2=1-\frac{\sum_{i=1}^{n}(y_i-\hat{y}_i)^2}{\sum_{i=1}^{n}(y_i-\overline{y})^2} \quad (6)$$

$$RMSE=\sqrt{\frac{\sum_{i=1}^{n}(y_i-\hat{y}_i)^2}{n}} \quad (7)$$

where y_i is the actual observed value, $\hat{y}_i$ is the predicted value, $\hat{y}$ is the average of the actual observed value, and n is the sample size.

3. Results

3.1. Physiological parameter statistics and analysis of variance

The statistical analysis of the data collected from two planting seasons was conducted for dry matter content, SPAD, plant height, and bending resistance force. As shown in Fig. 6-A, there were significant differences in the dry matter content measurements of the N1 treatment without nitrogen application compared to the other treatments. Most of the differences in dry matter content among the other treatments were not significant. As indicated in Fig. 6-B, significant differences in ear dry matter content occurred between H2, H6 and H1, H3, H4, H7, and in stem dry matter content between H2, H3, H5, H6 and H4, H7. Total dry matter content significantly differed between H5, H6, H7 and H4, H8. These results showed that differences among varieties were inconsistent across plant parts. The plant height, SPAD value and bending resistance force of the N1 treatment differed significantly from those of the other treatments, while there were no significant differences among the other treatments. Based on the data in Table 3, the bending resistance force, SPAD value and total dry matter content showed significant differences among the varieties, and there were highly significant differences in leaf, stem and total dry matter, and SPAD value among the experimental treatments. In short, a large amount of differentiated physiological parameter data was created through the field experiments with different treatments, which allowed the building of an inversion model with stronger generalization capabilities.

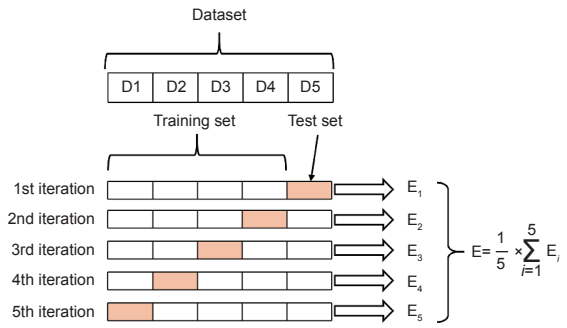


Fig. 5 Five-fold cross-validation schematic diagram.

3.2. Correlations between the physiological parameters

The heatmap in Fig. 7 was generated by calculating the Pearson correlation coefficients between physiological parameters and features extracted from the remote sensing images. The color represents the Pearson correlation coefficient, where red indicates a positive correlation, blue indicates a negative correlation, and deeper colors signify higher correlations. In terms of physiological parameters, plant height exhibits high correlations with all features. The ear dry matter content shows a moderate correlation with the mean of local binary pattern texture features and spectral features around 600 nm. Leaf, stem, and total dry matter contents exhibit high correlations with the reflectance values of the multispectral REG and NIR bands. Bending resistance force is highly correlated with the grayscale co-occurrence matrix of texture features and image entropy, while SPAD does not show any noticeably highly correlated features.

The distribution of Pearson correlation coefficients between different band values and texture features, as well as physiological parameters, were also calculated and compared before and after reconstruction, and the results are shown in Fig. 8. Except for ear dry matter content and SPAD, the Pearson correlation coefficients for most of the physiological parameters were generally higher before reconstruction. In addition, Pearson correlation coefficients for these two parameters were mostly below 0.5. Based on the correlation analysis, the correlations between the hyperspectral images reconstructed later and the six physiological parameters involved in the inversion in this study were not higher than that before reconstruction.

3.3. Parameter inversion of the individual features

In this study, six regression algorithms were implemented as described in Section 2.6 to establish regression models between each feature and physiological parameter. Five-fold cross-validation was employed to evaluate the accuracy of the predictions. The prediction accuracies of the six models were shown in Fig. 9. Overall, decision trees, linear neural networks, and multi-layer perceptron were not suitable for fitting the small sample dataset in this study, while the linear regression and polynomial regression models performed

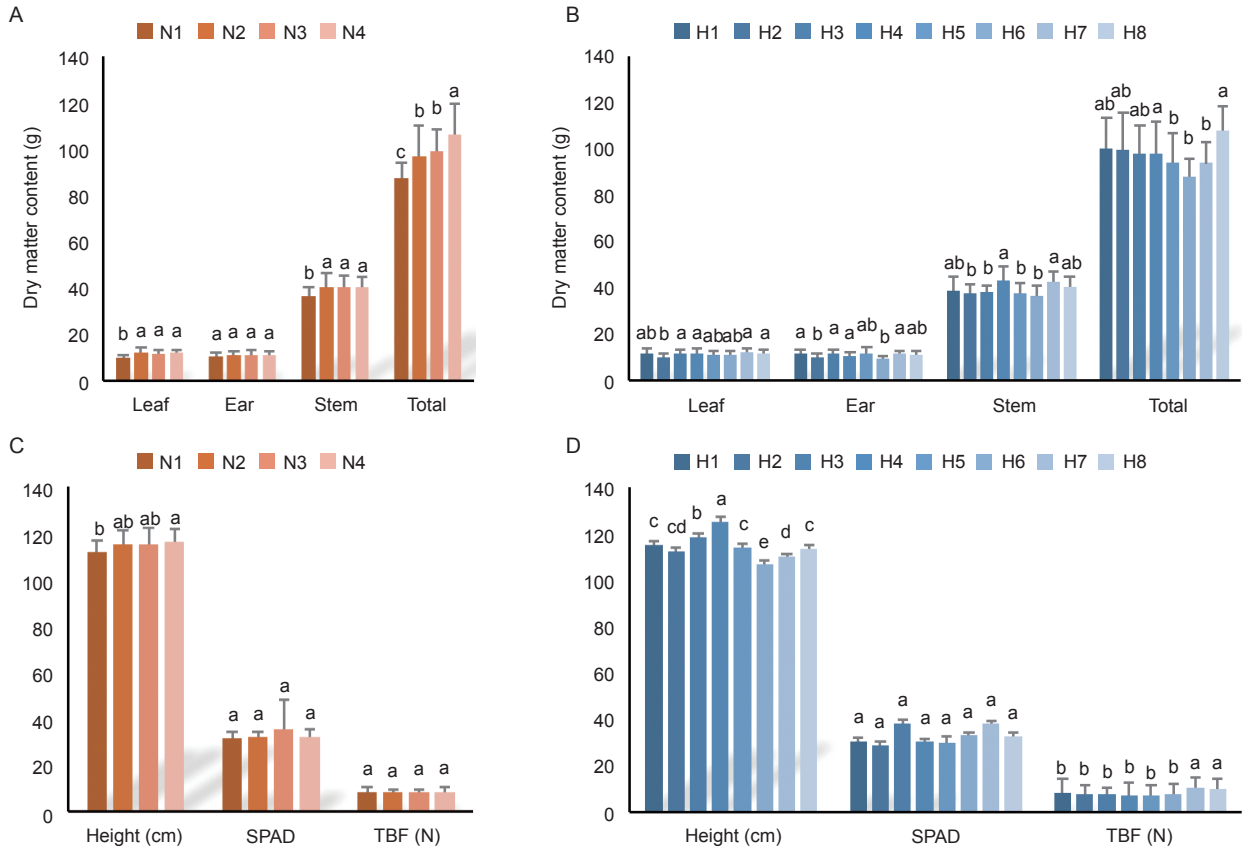


Fig. 6 Statistics of dry matter content, plant height, SPAD, and the bending resistance force (TBF) for different treatments and varieties. N1 to N4, total nitrogen application amounts of 0, 75, 105, and 135 kg ha^{-1}, respectively. H1 to H8, Meizhanxiang 2, Xiangyaxing, 19 Xiang, Ruanhua, Qingxiangyou 033, Nanjingxiang, Erguangxiang 3, and Lixiangzhan, respectively. Letters mean significant at the level of 0.05.

Table 3 Analysis of variance of the different physiological parameters in different varieties and treatments

Physiological parameters[1)]	Statistical index[2)]	Variety	Treatment	Variety× Treatment
Dry matter content				
Leaf	F	1.348	28.224	0.588
	P	0.231	0.000**	0.922
Ear	F	1.921	2.636	0.743
	P	0.069*	0.052*	0.783
Stem	F	5.454	14.521	1.064
	P	0.000**	0.000**	0.392
Total	F	1.092	8.269	0.735
	P	0.371	0.000**	0.792
Height	F	1.639	1.538	0.042
	P	0.128	0.207	1.000
SPAD	F	10.505	4.721	1.034
	P	0.000**	0.003**	0.426
TBF	F	5.083	2.623	0.693
	P	0.000**	0.052*	0.836

[1)] TBF, the bending resistance force.
[2)] F, *F*-value from ANOVA; P, *P*-value (significance level).
* and **, significance levels at 0.05 and 0.01, respectively.

better. In terms of physiological parameters, stem height exhibited the best prediction performance. The models constructed with data before and after reconstruction performed better than those for other parameters.

Combining the results in Fig. 10, the overall accuracy of the spectral data after reconstruction was generally not higher than that before reconstruction. However, as shown in Fig. 10-A–D, the maximum values of SPAD and bending resistance force were higher for the models constructed with data after reconstruction, and the dry matter content of the ear exhibited better performance in the models built with data after reconstruction. However, the accuracy of inversion of these physiological parameters was not high. In addition, the stem height inversion model constructed using the texture feature had a superior effect.

To determine the optimal inversion model for each physiological parameter, the model with the highest inversion accuracy for each physiological parameter was identified by sorting, as shown in Fig. 11. In addition to ear weight, other dry matter contents and plant heights were better inferred using the original data. The stem and total dry matter content showed improved performance when using the minimum value of the blue band in the experimental area. The dry matter content of the leaf exhibited the best inversion effect using the mean of the near-infrared band, and the stem height inversion employing the contrast of the grayscale co-occurrence matrix had the best effect, achieving a determination coefficient of 0.87. The ear dry matter content, SPAD, and bending resistance force yielded better results when using the spectral data after reconstruction.

3.4. Parameter inversion of the multiple features

To investigate whether the accuracy of physiological parameter inversion would be improved through modeling with combined features, six different feature combination methods were studied and each method was divided into four types, as shown in Fig. 12. For spectral features, the four types were the mean value, maximum value, minimum value, and standard deviation of the experimental area. For texture features, the four types were grayscale co-occurrence matrix, local binary pattern, image entropy, and all texture

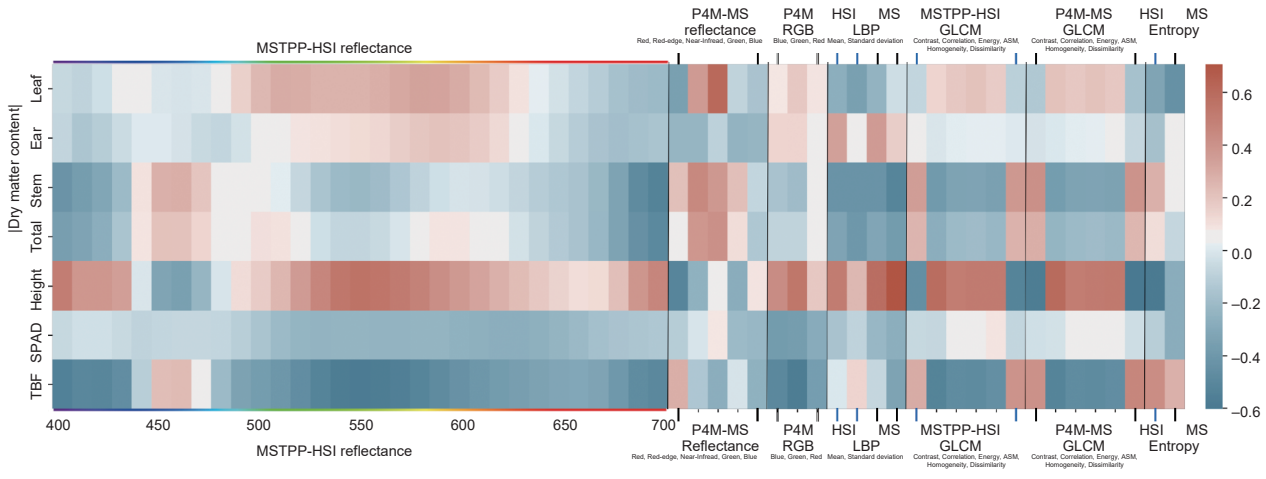


Fig. 7 Pearson correlation coefficient heatmap for analyzing the correlations between physiological parameters and image features. MS, multispectral image; HSI, hyperspectral image. Red, green, and other bands represent the band value of the image. GLCM, gray-level co-occurrence matrix; LBP, local binary patterns; Entropy, the texture features of an image. MSTPP, multistage spectral-wise transformer; P4M, phantom 4 multispectral.

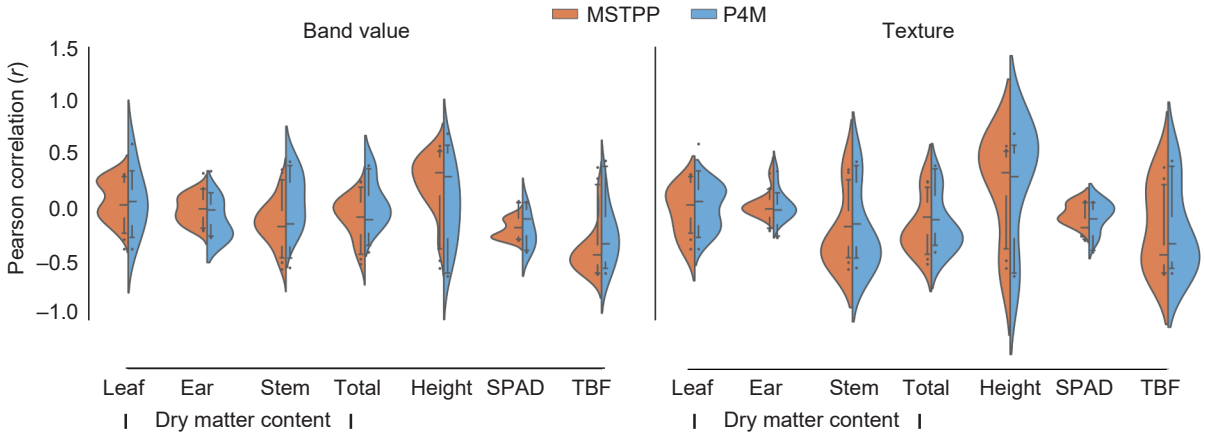


Fig. 8 Distribution of correlations between different types of features and physiological parameters before and after reconstruction. The outlier range was set to [5%, 95%]. MSTPP, multistage spectral-wise transformer; P4M, phantom 4 multispectral. TBF, the bending resistance force.

features. Combinations A and B were respectively the reconstructed full-spectral band features and texture features, while combinations C, D, and E were respectively the full spectral band, five single-channel multispectral bands, and three visible light spectral bands before reconstruction, and combination F was the texture features before reconstruction. Overall, the prediction accuracy of stem height remained the highest among the various feature combinations, and the combination of features did not improve the construction of a multi-layer perceptron inverse model.

As shown in Fig. 13, the feature combinations after reconstruction performed worse than those before reconstruction in most models and physiological parameter inversions, except for the ear and total dry matter contents, which were significantly better than the combinations before reconstruction (Fig. 13-C).

To determine the best inversion model for each physiological parameter after feature combination, the models with the highest accuracy for each physiological parameter inversion were identified through sorting (Fig. 14). Compared to before the combination, the accuracy of dry matter content inversion improved to some extent, and the best prediction accuracy of total dry matter content changed from using the best prediction accuracy of a single feature before reconstruction to employing the data modeling prediction accuracy after reconstruction. However, the best inversion models for SPAD and bending resistance force after feature combination showed reduced prediction accuracy, and the best models used data that had changed from before reconstruction to after reconstruction.

3.5. Reconstruction validity and model selection

The inversion model was constructed by combining different features, and the inversion effect of hyperspectral reconstruction on rice physiological parameters was found to be not as good as that of using multispectral images. However, since the reconstruction input was visible light images, the inversion results must be compared with those of the visible light images to evaluate whether the reconstruction can improve the inversion model. The accuracy of the inversion model constructed using reconstructed images was higher than that constructed using RGB images, except for the content of leaf dry matter (Fig. 15). By comparing the best models of the visible light, multispectral and reconstructed hyperspectral images for the inversion of rice physiological parameters (Table 4), reconstruction did not show a general improvement effect on the inversion model constructed by multispectral images, but it had a general improvement effect on the model constructed by visible light images, although that improvement effect differed for the different parameters.

The selection of appropriate machine learning models is helpful for improving inversion accuracy. This study compared the distribution of determination coefficients of all the inversion models described above, and set the data other than 0.05 to 0.95 as outliers, which are represented by the points in Fig. 16. Clearly, the inversion accuracies of the six selected machine learning models in this study had great differences for each parameter. The three machine learning models of linear regression, polynomial regression and support vector machine regression performed relatively well, and the determination coefficients of most models were greater than 0, while the inversion models constructed by the other three machine learning models of decision tree regression, linear neural network and multi-layer perceptron were less accurate.

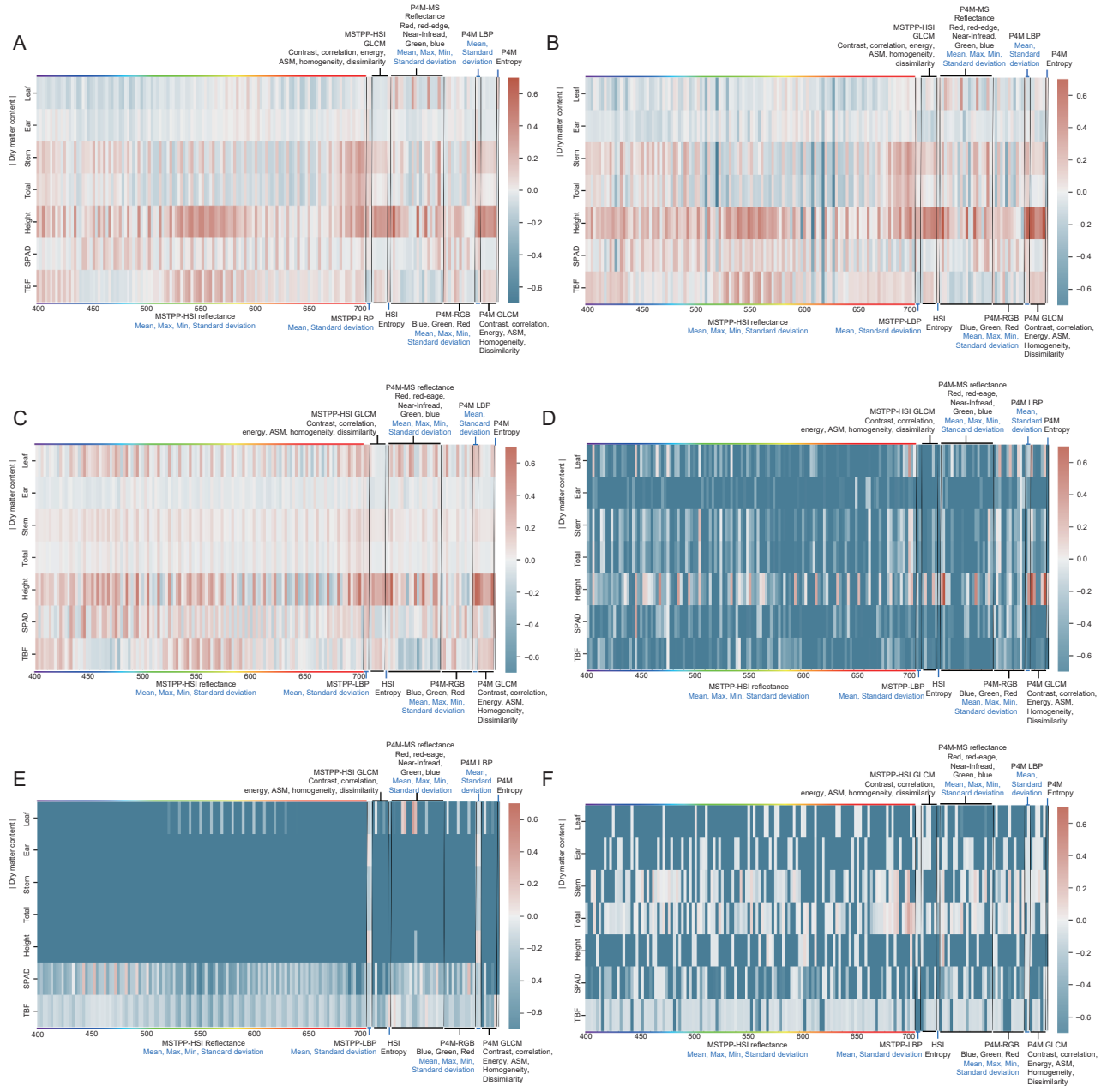


Fig. 9 Assessment of the six models for prediction accuracy. A, R^2 values of the linear regression model. B, R^2 values of the polynomial regression model. C, R^2 values of the support vector machine regression model. D, R^2 values of the decision tree regression model. E, R^2 values of the linear neural network regression model. F, R^2 values of the multilayer perceptron regression model. MS, multispectral image; HSI, hyperspectral image. Red, Green, and other bands represent the band value of the image. GLCM, gray-level co-occurrence matrix; LBP, local binary patterns; Entropy, the texture features of an image. MSTPP, multistage spectral-wise transformer; P4M, phantom 4 multispectral.

4. Discussion

4.1. Effectiveness of hyperspectral reconstruction technology

In the traditional task of inverting crop physiological parameters, the use of hyperspectral data, vegetation indices, and texture features can improve the accuracy of the inversion process (Al-Saddik *et al.* 2018; Yu *et al.* 2018; Han *et al.* 2022; Liu *et al.* 2023). The inversion accuracies for plant height, LAI and some directly observable parameters are relatively high in most studies. The RMSE percentages (the percentage of the ratio of the RMSE of the predicted value to the mean of the observed

value) for parameters such as dry matter and SPAD are generally around 5 to 10%. The accuracy of the inversion model used in this study was essentially equivalent to that of the traditional inversion method. Moreover, to more comprehensively evaluate the impact of hyperspectral reconstruction technology on inversion accuracy, this study used spectral values, texture features and various machine learning algorithms.

Hyperspectral reconstruction technology is an important data mining technique in the remote sensing field that has the potential to achieve low-cost hyperspectral data acquisition and significantly impact the promotion and application of remote sensing technology. This study inferred the reconstructed remote sensing data based on the MST++ hyperspectral reconstruction model, combined it with a large amount of data collected in the experimental area, and used various algorithms to construct inversion models. By comparing and exploring these options, we investigated whether a hyperspectral reconstruction model can facilitate the remote sensing inversion task. The results show that hyperspectral reconstruction does not universally improve the accuracy of the inversion task when multispectral sensors serve as the data source (Figs. 10 and 13).

4.2. Comparative analysis of single and multi-feature modeling

For single feature modeling (Fig. 10), the reconstructed data were more effective for the inversion model of rice stem, ear, and total dry matter contents, as well as SPAD and bending resistance force. The 95% quantiles were always higher than the original data in the various modeling methods. However, the reconstructed data did not work well for plant height or leaf dry matter content. In particular, the texture features of the original data were more effective for the plant height inversion model. This indicates that physiological parameters which can be directly observed are more suitable when using the original multispectral data to construct the inversion model, while physiological parameters that are not easily observed require more complex feature extraction methods to improve the accuracy of the model.

For the inversion model based on multi-feature combinations (Fig. 13), the reconstructed data had only a certain advantage in estimating the dry matter content of the panicle. However, the inversion accuracies of SPAD and TBF based on multi-feature combination declined. This indicated that multi-feature combination can replace

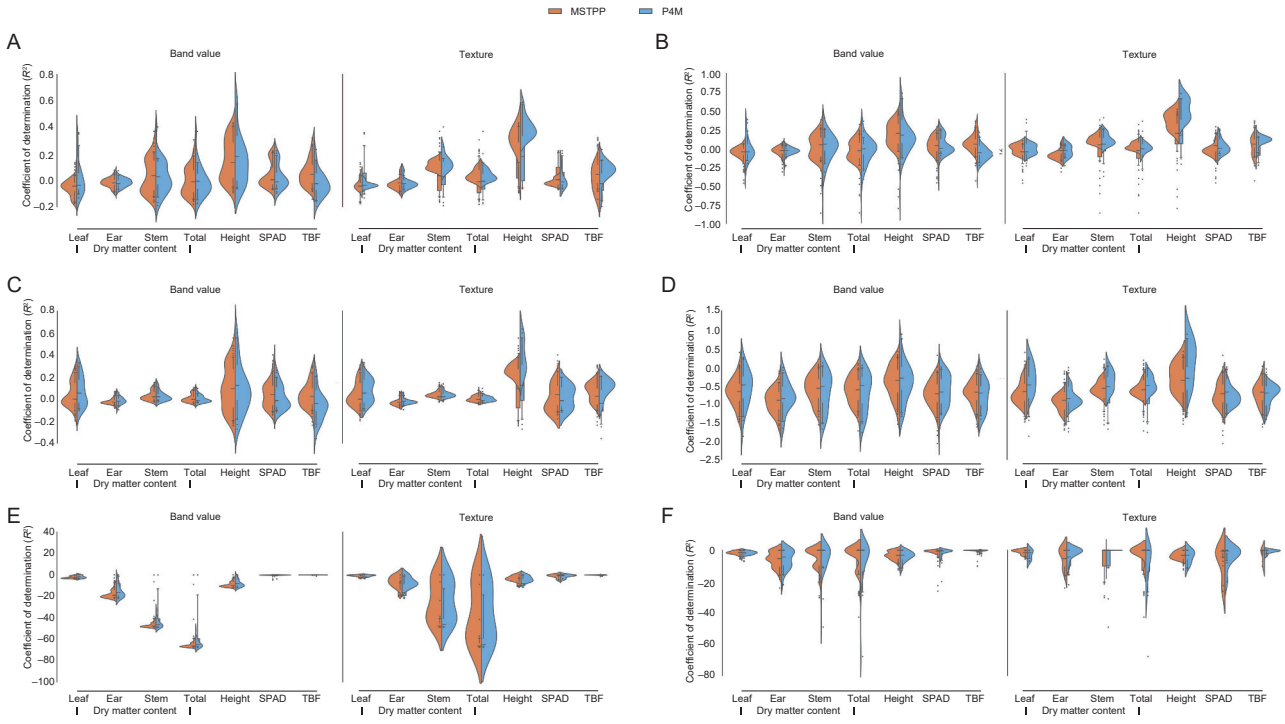


Fig. 10 Distribution of the inversion model accuracies of six models for different types of features and physiological parameters before and after reconstruction. The outlier range was set to [5%, 95%]. A, distribution of R^2 values of the linear regression model. B, distribution of R^2 values of the polynomial regression model. C, distribution of R^2 values of the support vector machine regression model. D, distribution of R^2 values of the decision tree regression model. E, distribution of R^2 values of the linear neural network regression model. F, distribution of R^2 values of the multilayer perceptron regression model. MSTPP, multistage spectral-wise transformer; P4M, phantom 4 multispectral. TBF, the bending resistance force.

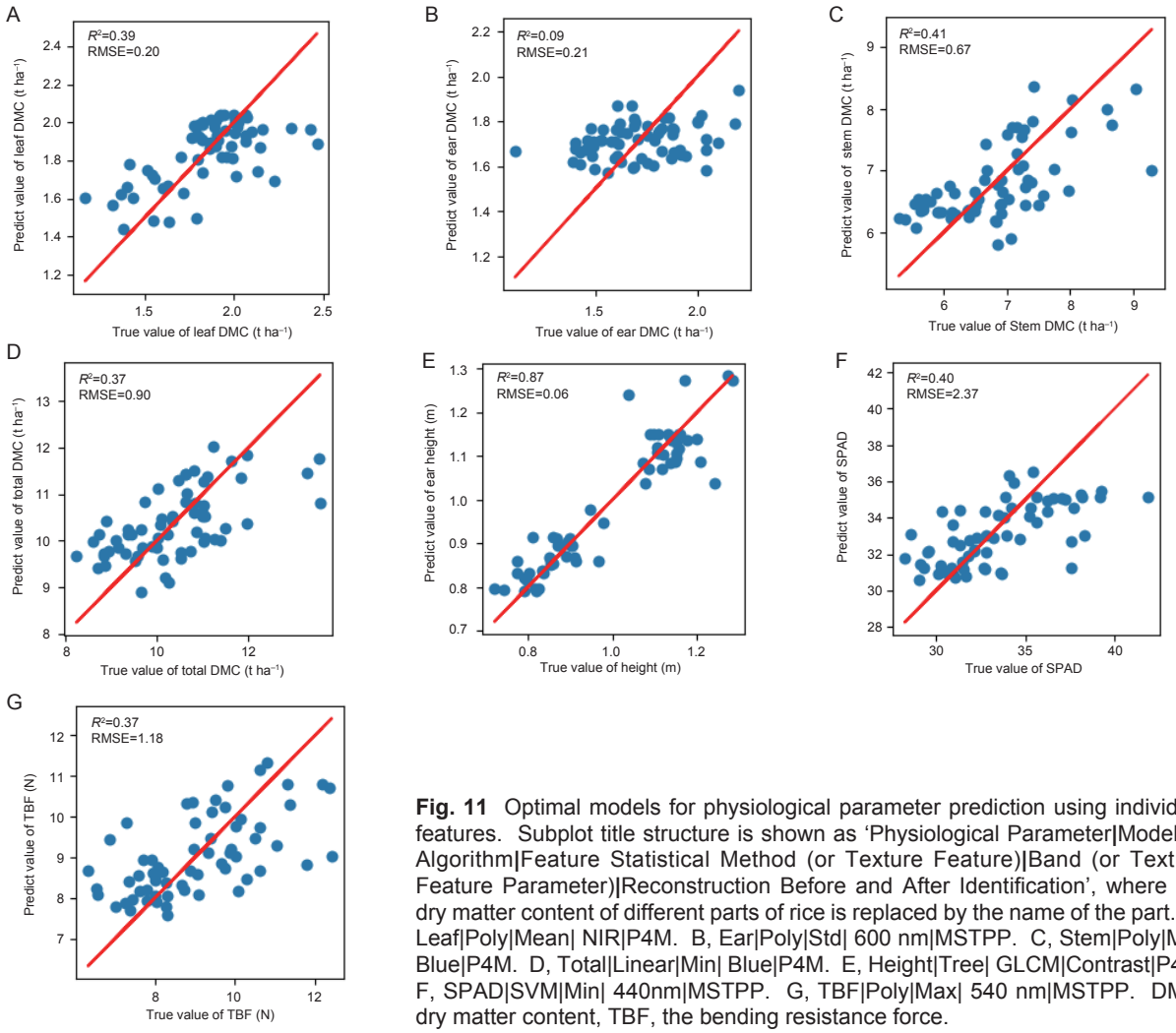
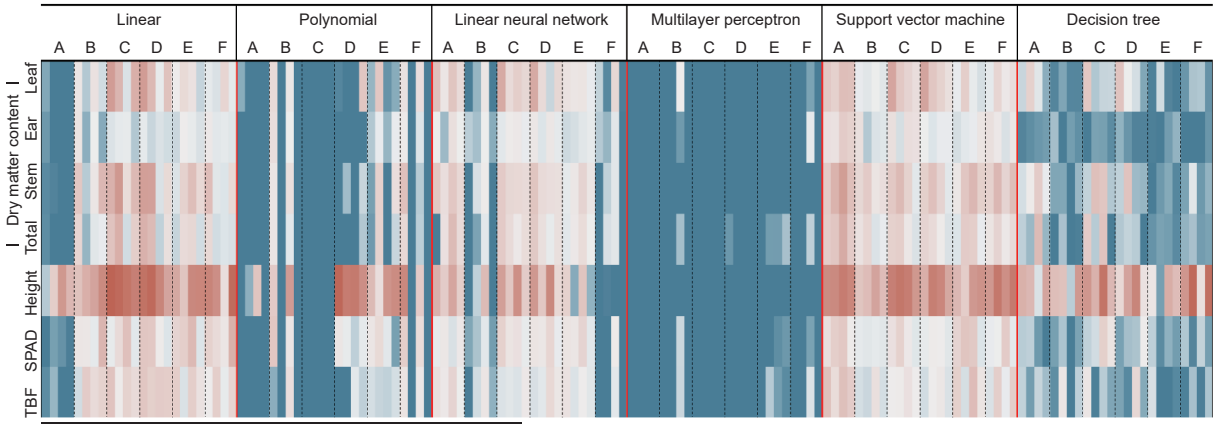


Fig. 11 Optimal models for physiological parameter prediction using individual features. Subplot title structure is shown as 'Physiological Parameter|Modeling Algorithm|Feature Statistical Method (or Texture Feature)|Band (or Texture Feature Parameter)|Reconstruction Before and After Identification', where the dry matter content of different parts of rice is replaced by the name of the part. A, Leaf|Poly|Mean| NIR|P4M. B, Ear|Poly|Std| 600 nm|MSTPP. C, Stem|Poly|Min| Blue|P4M. D, Total|Linear|Min| Blue|P4M. E, Height|Tree| GLCM|Contrast|P4M. F, SPAD|SVM|Min| 440nm|MSTPP. G, TBF|Poly|Max| 540 nm|MSTPP. DMC, dry matter content, TBF, the bending resistance force.



A	MSTPP band value	Mean, Max, Min, Standard deviation
B	MSTPP texture	LBP, GLCM, entroy, All texture
C	P4MALL band value	Mean, Max, Min, Standard deviation
D	P4M MS band value	Mean, Max, Min, Standard deviation
E	P4M RGB band value	Mean, Max, Min, Standard deviation
F	P4M texture	LBP, GLCM, entroy, All texture

Fig. 12 Pearson correlation coefficient heatmap for analyzing the correlations between physiological parameters and image features. MSTPP, multistage spectral-wise transformer; P4M, phantom 4 multispectral. TBF, the bending resistance force.

feature mining in the reconstruction process to a certain extent, but the inversion accuracy of physiological parameters that cannot be directly observed is limited when using only original data.

4.3. Challenges and limitations of hyperspectral data reconstruction

The results indicated that compared to the visible light image that was previously used for hyperspectral reconstruction, the reconstructed data exhibited a significant improvement in the correlations with physiological parameters (Fig. 7). The accuracy improved in both the single-feature (Fig. 9) and combined feature inversion models (Fig. 15). This occurred because the hyperspectral reconstruction data introduces a larger amount of spectral information compared to the visible light image, and the information mining in the visible light image is more thorough. Therefore, for remote sensing data inversion tasks using visible light sensors as the source, implementing hyperspectral reconstruction technology can significantly improve accuracy.

This study showed that hyperspectral reconstruction has a certain practical value in remote sensing inversion tasks; however, the reconstruction model based on visible light may not be able to meet the high-precision inversion requirements. The MST++ model used the dataset provided by the NTIRE 2022 Spectral Reconstruction Challenge in its model training, which contains 1,000 pairs of RGB to hyperspectral data (Cai *et al.* 2022). However, this dataset lacks agricultural settings, which may be the reason why the reconstructed data did not significantly improve the accuracy of inversion modeling. In the future, more hyperspectral data from agricultural scenarios will be necessary to enhance the hyperspectral reconstruction model.

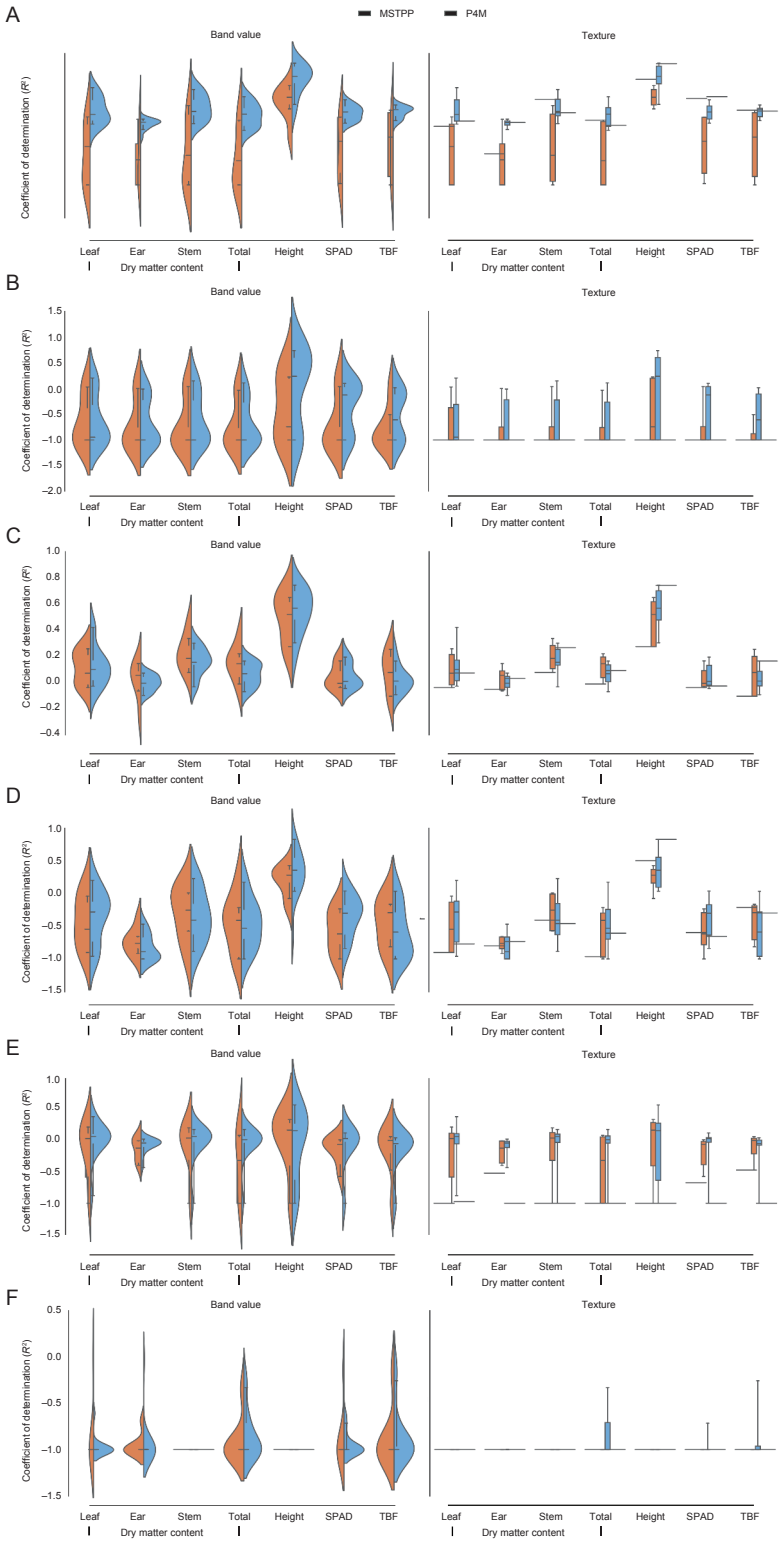


Fig. 13 Distribution of modeling accuracies for six models under multiple feature combinations before and after reconstruction. The data range was set to [–1,1], and the outlier range was set to [5%, 95%]. A, distribution of R^2 values of the linear regression model. B, distribution of R^2 values of the polynomial regression model. C, distribution of R^2 values of the support vector machine regression model. D, distribution of R^2 values of the decision tree regression model. E, distribution of R^2 values of the linear neural network regression model. F, distribution of R^2 values of the multilayer perceptron regression model. MSTPP, multistage spectral-wise transformer; P4M, phantom 4 multispectral. TBF, the bending resistance force. Bars indicate 5th–95th percentiles (n=174).

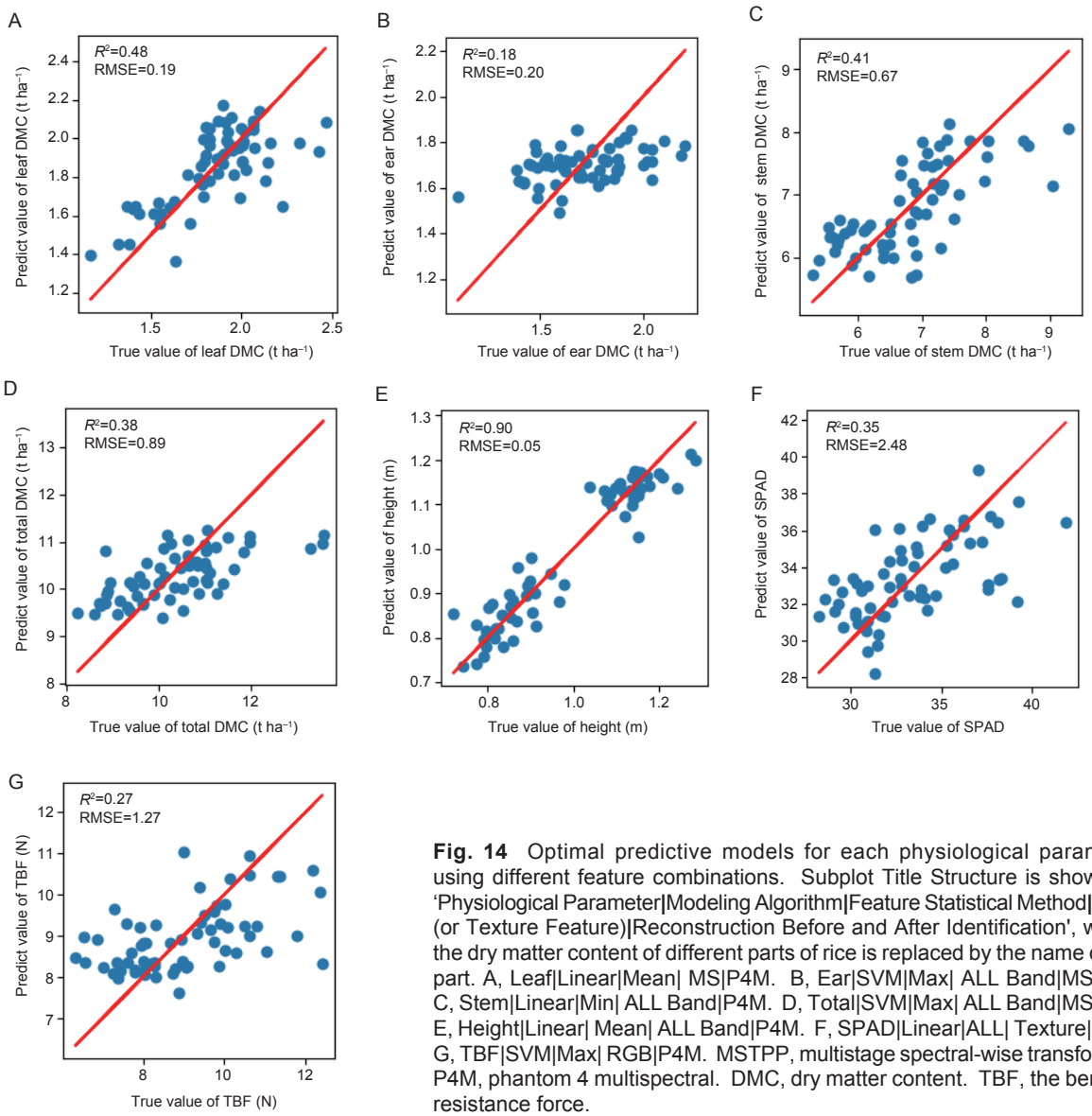


Fig. 14 Optimal predictive models for each physiological parameter using different feature combinations. Subplot Title Structure is shown as 'Physiological Parameter|Modeling Algorithm|Feature Statistical Method|Band (or Texture Feature)|Reconstruction Before and After Identification', where the dry matter content of different parts of rice is replaced by the name of the part. A, Leaf|Linear|Mean| MS|P4M. B, Ear|SVM|Max| ALL Band|MSTPP. C, Stem|Linear|Min| ALL Band|P4M. D, Total|SVM|Max| ALL Band|MSTPP. E, Height|Linear| Mean| ALL Band|P4M. F, SPAD|Linear|ALL| Texture|P4M. G, TBF|SVM|Max| RGB|P4M. MSTPP, multistage spectral-wise transformer; P4M, phantom 4 multispectral. DMC, dry matter content. TBF, the bending resistance force.

Furthermore, as the dataset only contains 31 bands in the range of 400–700 nm, the lack of infrared bands greatly limits its application in agricultural remote sensing, and a hyperspectral reconstruction model suitable for infrared bands will be developed in the future.

4.4. Machine learning models in inversion accuracy

This study used six machine learning models, including two artificial neural network models based on backpropagation. The inversion accuracies of the two neural network models and the decision tree regression model were poor. In addition, finding the optimal solution is difficult when the dataset is small, regardless of whether it is a neural network or a decision tree, as discussed in previous studies (Yuan *et al.* 2017; Nagasubramanian *et al.* 2018). However, the 960 rice plants collected were grouped into 64 sets of datasets by measuring their physiological parameters and statistics according to the test area, which is far from sufficient for the neural network models. Nevertheless, note that for the construction of small-sample inversion models, simple linear regression can still achieve a good fitting effect, as shown in Figs. 14-E and 16. Therefore, choosing the appropriate machine learning models is crucial for building the inversion model.

4.5. Future directions and research needs

The current study did not account for the inversion effect of infrared band reconstruction, and did not verify the conclusion or test the generalizability of the model in multiple regions or planting seasons. Therefore,

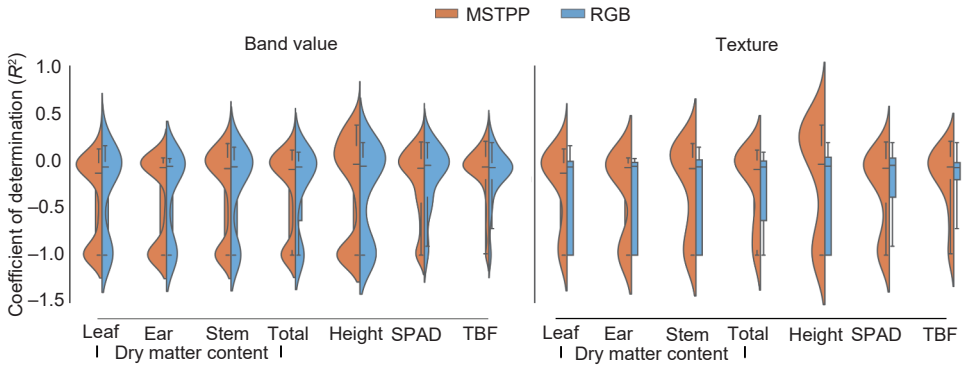


Fig. 15 Distribution of prediction accuracies for the visible light and inverse models after reconstruction. MSTPP, multistage spectral-wise transformer; RGB, visible light. TBF, the bending resistance force.

Table 4 R^2 of the best inversion model under different data sources and its enhancement ratio by hyperspectral reconstruction

Parameter[1)]	Max R^2 [2)]			Enhancement ratio (%)[3)]	
	MSTPP	P4M	RGB	P4M	RGB
Dry matter content					
Leaf	0.38	0.48	0.32	–21.36	19.03
Ear	0.18	0.08	0.04	130.21	392.42
Stem	0.47	0.50	0.19	–6.28	143.98
Total	0.38	0.37	0.14	3.74	166.98
Height	0.71	0.90	0.25	–21.12	183.57
SPAD	0.40	0.35	0.26	16.16	53.10
TBF	0.37	0.27	0.27	34.78	37.92

[1)] TBF, the bending resistance force.
[2)] MSTPP, multistage spectral-wise transformer; P4M, phantom 4 multispectral. RGB, visible light image.
[3)] The enhancement ratio was calculated by the formula: (R^2_{MSTPP}–$R^2_{(P4M/RGB/R2(P4M/RGB)}$), where R^2 is that of the best model under the current data type.

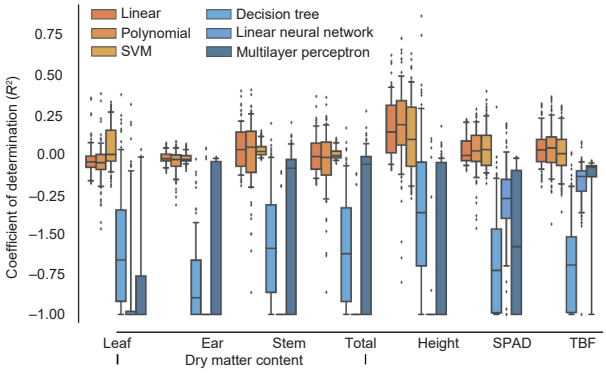


Fig. 16 Prediction accuracies of the different algorithms. TBF, the bending resistance force. SVM, support vector machine.

future research should focus on whether hyperspectral reconstruction technology that includes the infrared band is useful for the inversion of physiological parameters. In addition, it is necessary to explore whether the small-sample model in the inversion field can be generalized.

5. Conclusion

This study employed the MST++ hyperspectral reconstruction model and several machine learning algorithms to reconstruct remote sensing images from two rice-growing seasons in 2021, comparing inversion accuracies before and after reconstruction. Results indicate that hyperspectral reconstruction can significantly enhance model performance when using visible-light data; for instance, the highest R^2 improvement for plant height was over 180%. However, it does not universally surpass multispectral data performance — plant height inversion accuracy decreased by over 20% compared to original multispectral data. Specifically, for directly observable parameters (e.g., plant height, leaf traits), original multispectral data yielded better inversion models, outperforming reconstructed data by more than 20% in R^2. Conversely, for indirectly observable parameters (e.g., ear and total dry matter content, SPAD, TBF), hyperspectral reconstruction enhanced inversion accuracy, with R^2 improvements ranging from 3 to 130%. Therefore, traditional agronomic plot experiments should prioritize appropriate feature extraction and simpler regression methods to achieve high accuracy. Future hyperspectral reconstruction advancements should target broader wavelengths and diverse agricultural scenarios.

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Declaration of competing interest

The authors declare that they have no conflict of interest.

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